DocPolicyKG: A Lightweight LLM-Based Framework for Knowledge Graph Construction from Chinese Policy Documents

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Abstract

Chinese policy documents are typically written in a concise yet contextually rich style, containing implicit hierarchical logic and strategic intent. These linguistic and structural features present challenges for traditional information extraction methods, which often struggle with cross-sentence dependencies and semantic complexity. To address these, we propose DocPolicyKG, a novel framework for constructing knowledge graphs from Chinese policy documents using lightweight large language models (LLMs), integrating domain ontology, fine-tuning, and prompt engineering. Focusing on the investment promotion policies in China, experimental results demonstrate that DocPolicyKG significantly outperforms the base model Deepseek-R1-7B and achieves performance competitive with GPT-4o on both named entity recognition (NER) and relation triplet extraction (RTE) tasks. Building on DocPolicyKG, we construct the first large-scale knowledge graph of Chinese investment promotion policies, and further integrate Graph Retrieval-Augmented Generation (Graph RAG) to support policy question answering through entity-relation reasoning and semantic retrieval. The graph data is publicly available at: `github.com/hanshenmesen/DocPolicyKG`.

CCS Concepts

• **Computing methodologies** → **Information extraction**.

Keywords

Chinese Policy Texts, Document-level Information Extraction, Knowledge Graph Construction, Large Language Models, Graph RAG

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1 Introduction

Chinese policy documents exhibit distinctive linguistic and structural features, including concise expression, implicit hierarchical logic, and rich contextual framing [18]. While these documents serve as an essential medium for conveying governance intentions and guiding administrative implementation, their unstructured formats and intricate semantics present challenges for automated information extraction [3]. Traditional approaches, such as rule-based matching [22] and statistical learning methods [10], often fail to process such texts effectively due to their underlying semantic complexity and cross-sentence dependencies [11]. Recent advances in large language models (LLMs) offer new possibilities for addressing these challenges. Despite their strong contextual understanding, relying on powerful models remains costly, and they often struggle with the complex structure of Chinese policy texts [3].

In this study, we propose DocPolicyKG, a novel framework for constructing domain-specific knowledge graphs from Chinese policy documents using lightweight LLMs. The framework integrates ontology design [5], parameter-efficient fine-tuning [9], and prompt engineering [12, 13] to support both Named Entity Recognition (NER) and Relation Triplet Extraction (RTE). By explicitly modeling cross-sentence dependencies and aligning with the semantic characteristics of Chinese policy discourse, DocPolicyKG achieves high extraction accuracy while maintaining computational efficiency.

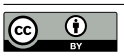
We apply DocPolicyKG to the domain of investment promotion, a policy area with strategic importance for capital attraction and regional development in China [7, 16]. Experimental results on annotated datasets show that DocPolicyKG significantly outperforms the base model Deepseek-R1-7B [1] and achieves performance comparable to GPT-4o. Leveraging 3,440 policy documents from 1994 to 2024, we construct the first knowledge graph in this domain, comprising 16,618 entities and 52,596 relations. Besides, the Graph Retrieval-Augmented Generation (Graph RAG) [6, 23] framework is adopted to support more accurate and interpretable policy question answering with structured knowledge.

Our main contributions are summarized as follows:

(1) We propose DocPolicyKG, a lightweight LLM-based framework for document-level information extraction from Chinese policy texts.

(2) We construct the first large-scale knowledge graph for investment promotion policies in China.

(3) We integrate the knowledge graph with Graph RAG to support contextually grounded policy question answering.



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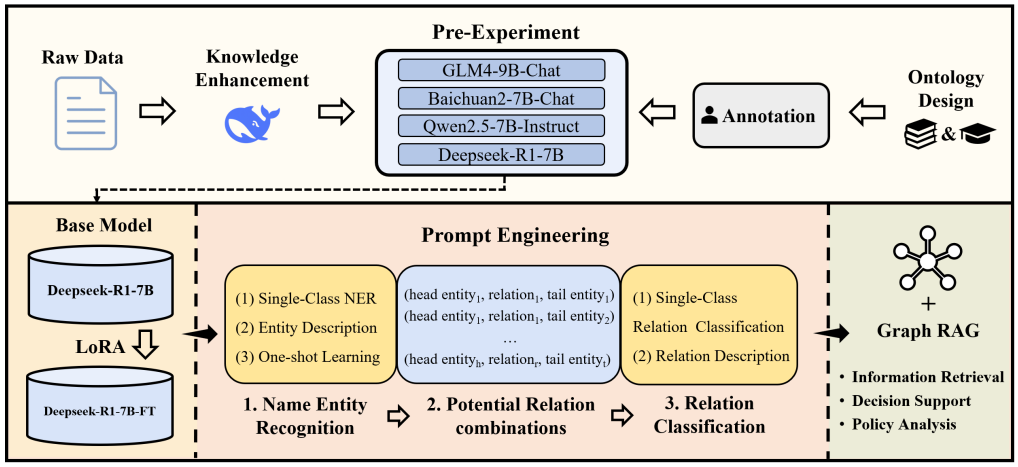


Figure 1: Framework of the DocPolicyKG framework.

2 Our Framework: DocPolicyKG

2.1 Ontology Design

DocPolicyKG is grounded in a structured domain ontology developed through consultation with domain experts and systematic analysis of Chinese policy documents [5, 18]. Given the linguistic characteristics of these documents, such as implicit subject references, layered administrative logic, and abstract expressions of intent, the ontology is specifically tailored to accommodate the semantic conventions prevalent in Chinese governance discourse.

Tailored to the domain of investment promotion, the ontology defines core entity and relation types that capture the semantic patterns of Chinese policy discourse, including administrative actors, documents, locations, and policy intents, as summarized in Table 1. This ontology-driven design supports scalable knowledge graph construction while preserving hierarchical and contextual structures in the texts.

2.2 Data and Knowledge Enhancement

We construct a corpus of 3,440 investment promotion policy documents issued between 1994 and 2024, retrieved from the Peking University Law Database[1] by filtering titles containing the keyword "investment promotion." All texts are preprocessed through format normalization, date unification, and noise removal.

To improve the extraction quality from Chinese texts, we apply the knowledge enhancement process with Deepseek-R1 (API). The prompt consists of three steps: (1) enhancing linguistic clarity and structural coherence while preserving semantics; (2) removing appendices and redundant content; and (3) replacing vague references with explicit entity mentions. The refined output exhibits improved lexical consistency and semantic specificity, as demonstrated in our open repository. In the absence of public benchmarks in this domain, a custom annotated corpus is developed using the Doccano platform. Documents are selected in proportion to their annual distribution and a total of 300 documents are annotated, As summarized in Table 1.

[1]https://www.pkulaw.com/

Table 1: Statistics of annotated entity and relation types.

Entity Type	Count	Relation Type	Count
Government Agency	291	issues	503
Policy Document	616	located in	160
Enterprise/Organization	271	cooperates with	230
Province	281	cites	1015
Region	283	aims at	540
Date	338	based on	317
Project	381	includes	641
Industry	563	issued on	635
Policy Objective	264	targets region	601
Guiding Principle	182	targets industry	1001
Policy Type	297	classified as	459
Promotion Type	441	promotes by	798

2.3 Base Model and Fine-Tuning

We conduct preliminary experiments to evaluate four lightweight open-source LLMs—GLM4-9B [4], Qwen2.5-7B [14], Bai-chuan2-7B [2], and Deepseek-R1-7B [1] on information extraction of Chinese policy documents. The evaluation follows a multi-granularity approach: (1) phrase-level entities (e.g., Government Agency, Policy Document) are matched using exact string comparison; (2) sentence-level entities (Policy Objective, Guiding Principle) are evaluated using cosine similarity with a threshold of 0.75 [17]; and (3) relation triplets are considered correct only when all components—head, tail, and relation—are accurately extracted.

Deepseek-R1-7B achieves the best performance among the evaluated models and is selected as the base model for subsequent experiments. To improve the quality of information extraction, Low-Rank Adaptation (LoRA) is employed to fine-tune the NER task with a training set of 1,000 samples generated by GPT-4o. No fine-tuning is conducted for the RTE task. No fine-tuning is conducted for the RTE task because it is reformulated as a classification task through prompt engineering.

Table 2: Results of the Named Entity Recognition.

Entity Type	Deepseek-R1-7B			DocPolicyKG			GPT-4o		
	F1	Recall	Precision	F1	Recall	Precision	F1	Recall	Precision
Government Agency	33.10	33.70	32.52	81.04	81.34	80.74	83.92	84.09	83.75
Policy Document	57.12	58.15	56.12	83.61	83.24	83.99	85.38	88.12	82.81
Enterprise/Organization	46.91	50.80	43.57	74.88	76.67	73.17	81.37	77.24	85.96
Province	75.78	74.05	77.59	92.07	90.27	93.94	94.38	96.41	92.44
Region	60.21	58.07	62.52	79.34	76.74	82.13	85.28	88.46	82.32
Date	77.19	77.93	76.47	90.82	90.85	90.79	96.62	99.42	93.98
Project	38.21	41.83	35.16	69.79	71.98	67.73	72.58	72.13	73.03
Industry	57.29	62.06	53.20	78.15	80.09	76.30	88.47	86.94	90.06
Policy Objective	43.20	48.08	39.22	75.57	73.60	77.65	77.48	72.58	83.09
Guiding Principle	42.65	40.14	45.49	72.55	69.30	76.11	74.29	77.73	71.14
Policy Type	72.16	75.00	69.53	86.86	89.54	84.34	88.51	88.83	88.20
Promotion Type	63.49	65.95	61.21	90.67	87.48	94.11	92.23	93.32	91.16
Overall	56.51	53.58	59.78	83.47	86.03	81.06	85.72	83.24	88.35

Table 3: Results of the Relation Triplet Extraction.

Relation Type	Deepseek-R1-7B			DocPolicyKG			GPT-4o			AutoRE*		
	F1	Recall	Precision	F1	Recall	Precision	F1	Recall	Precision	F1	Recall	Precision
issues	25.16	28.02	22.83	57.93	61.77	54.54	65.80	63.01	68.84	42.17	42.14	42.20
located in	35.10	30.79	40.81	80.76	76.97	84.94	87.16	90.34	84.19	39.48	34.79	45.63
cooperates with	7.29	5.55	10.63	48.67	46.04	51.62	56.19	53.90	58.68	48.21	46.59	49.95
cites	22.13	18.42	27.72	67.76	70.98	64.82	77.29	75.03	79.69	35.70	34.98	36.45
aims at	9.19	7.24	12.58	45.48	49.03	42.41	55.16	53.91	56.47	54.98	51.52	58.94
based on	33.98	35.00	33.02	66.83	68.53	65.21	69.38	70.45	68.34	42.14	46.37	38.61
includes	9.33	5.68	26.03	54.99	59.62	51.03	62.83	64.06	61.64	50.67	46.77	55.28
issued on	31.28	33.48	29.35	70.89	69.44	72.41	74.21	78.69	70.22	68.62	64.38	73.46
targets region	29.70	25.60	35.37	66.07	69.90	62.64	71.29	73.55	69.17	29.15	30.83	27.64
targets industry	16.98	12.86	24.99	64.36	67.90	61.17	70.21	68.21	72.33	51.04	50.98	51.10
classified as	36.10	40.78	32.38	78.79	76.46	81.26	86.91	82.95	91.27	65.13	62.88	67.54
promotes by	25.60	26.56	24.71	76.33	78.23	74.52	80.48	82.78	78.31	70.42	66.85	74.39
Overall	24.33	27.73	21.67	68.03	64.64	71.80	72.56	70.76	74.46	50.31	48.94	51.76

2.4 Prompt Engineering

The prompt for NER task is composed of five key components:

Role Definition: The model is prompted as a policy analysis expert familiar with the structure and terminology of investment promotion documents.

Task Specification: Each prompt targets a single entity type to reduce confusion, and requires structured JSON output [8].

Entity Definition: Clear definitions of the target entity type are provided to enhance extraction precision [21].

One-Shot Learning: To enable in-context learning, a representative example is included in the prompt. It is selected based on cosine similarity between the input text and samples from the fine-tuning set [15].

Input and Output: The model processes knowledge-enhanced policy texts and extracts all instances of the specified entity type.

For the RTE task, all possible head–tail entity pairs identified in the NER stage are combined according to the predefined relation schema and evaluated as binary classification task. This process **not only ensures comprehensive coverage of potential relations but also captures new triplets that may span across sentences and are missed in manual annotations.** The prompt includes the following four components:

Role Definition: The model is assigned the role of a policy analysis expert to ensure domain-consistent reasoning.

Task Formulation: Inspired by prior findings that LLMs perform better on classification tasks [19], each prompt evaluates whether a specified relation holds between a given head–tail entity pair within a policy document.

Relation Definition: Each relation type includes a concise definition of the target relation type to guide the model's judgment.

Input and Output: The input includes the policy text, head and tail entities, and a target relation. The model checks if the relation exists and outputs a binary result in JSON format.

3 Experiments

All experiments are conducted on an Ubuntu 20.04 system with an NVIDIA A100 GPU. The software environment includes Python 3.12, PyTorch 2.4, and CUDA 12.1. During fine-tuning, we use a learning rate of 2e-5, batch size of 8, and train for 8 epochs. To evaluate the effectiveness of DocPolicyKG, GPT-4o is adopted as a comparative baseline with the same prompts. For RTE, we include AutoRE [20] as an additional baseline and denote its prompt-only variant (no fine-tuning) as AutoRE*. Evaluation is based on precision, recall, and F1-score, with F1 as the main reported metric.

3.1 Named Entity Recognition

Table 2 reports the NER results. DocPolicyKG improves upon the Deepseek-R1-7B base model across all categories (+26.96), achieving an overall F1-score of 83.47. Notable gains are observed in *Government Agency* (81.04, +47.94) and *Policy Objective* (75.57, +32.17). GPT-4o achieves the best overall performance (85.72), especially

in structured types such as *Date* (96.62) and *Province* (94.38). However,DocPolicyKG demonstrates comparable performance in context-sensitive entities such as *Policy Objective* (77.48 vs. 75.57) and *Guiding Principle* (73.84 vs. 72.55), showing its strength in extracting complex entities while remaining efficient.

3.2 Relation Triplet Extraction

Table 3 reports the RTE results. DocPolicyKG achieves a substantial improvement over the base model Deepseek-R1-7B, raising the overall F1-score from 24.33 to 68.03. Significant gains are observed in both structured relations such as *Located in* (+45.66) and *Promotes by* (+50.73), as well as in context-dependent types like *Cooperates with* (+41.38) and *Aims at* (+36.29). GPT-4o achieves the best overall performance (72.56), consistently outperforming other models across all relation types. However, DocPolicyKG narrows the gap in several complex relations (e.g., *Includes*: 62.83 vs. 54.99; *Cites*: 77.29 vs. 67.76), demonstrating strong extraction capability under limited-resource settings.

These results reflect the effectiveness of DocPolicyKG in extracting knowledge from Chinese policy documents. While GPT-4o remains the best performance, DocPolicyKG provides a practical balance between accuracy and computational efficiency.

3.3 Ablation Analysis

We conduct ablation experiments to evaluate the contribution of key components in DocPolicyKG in Table 4:

Table 4: Ablation Analysis (F1 Scores).

Model Setting	NER	Δ	RTE	Δ
DocPolicyKG	83.47	–	68.03	–
w/o Model Fine-tuning	63.27	**↓20.20**	35.20	**↓32.83**
w/o Knowledge Enhancement	77.05	↓6.42	58.39	↓9.64
w/o Definition Descriptions	78.21	↓5.26	60.10	↓7.93
All Types Extracted Simultaneously	75.71	↓7.76	53.12	↓14.91
w/o In-Context Learning	72.26	**↓11.21**	–	–
Generative Instead of Discriminative	–	–	40.27	**↓27.76**

Removing fine-tuning results in significant performance drops of 20.20 (NER) and 32.83 (RTE), highlighting its necessity for domain adaptation. **Disabling knowledge enhancement** reduces F1-scores by 6.42 and 9.64, indicating the importance of contextual information. **Omitting entity/relation descriptions** causes further declines of 5.26 and 7.93, confirming the value of explicit prompt guidance. **Extracting all types simultaneously** results in performance drops of 7.76 (NER) and 14.91 (RTE), suggesting that type-wise separation helps reduce interference. In the setting **without in-context examples**, F1 decreases by 11.2, underscoring their role in domain-specific generalization. Last, **reframing RTE as a generation task** yields a 27.76-point drop, demonstrating the superiority of the classification-based formulation.

4 Knowledge Graph Construction

4.1 Graph Construction

We apply the DocPolicyKG to extract structured knowledge from 3,440 Chinese investment promotion policies. All extracted entities and relations are integrated into the Neo4j database, forming a knowledge graph comprising 16,618 entities and 52,596 relations. DocPolicyKG captures not only surface-level entities but also contextual relationships, including policy objectives, implementation pathways, and inter-document references. It supports entity-centric querying of structured attributes such as region, industry, and policy objective. Additionally, it enables compositional reasoning, such as retrieving provincial-level policies related to a specific sector and mapping hierarchical relationships across administrative levels.

4.2 Graph RAG Application

To evaluate the utility of the constructed knowledge graph for intelligent retrieval, we implement a Graph RAG system [6] based on the `nano-graphrag`[2] framework. Given a user query, the system generates a Cypher query to retrieve relevant entities and relations from Neo4j, and simultaneously retrieves semantically related document fragments from Milvu database. The results are then combined into the prompt to better guide the LLMs.

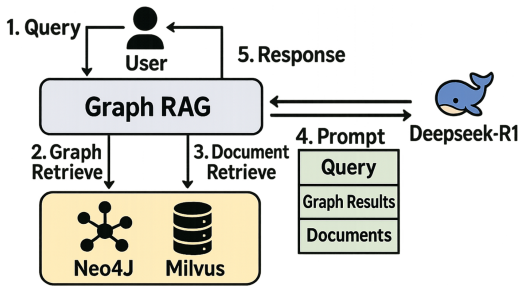


Figure 2: Architecture of Graph RAG, combining Neo4j-based graph retrieval with vector-based semantic retrieval.

For example, when asked "How has the AI industry developed in Anhui Province in recent years?", zero-shot strategy may hallucinate or miss key policy elements, and Native RAG may retrieve unnecessary fragments; In contrast, Graph RAG grounds its response in a structured relation chain *Anhui Province→AI Industry→Policy Document* and accurately identifies relevant policies and objectives, generating contextually grounded responses aligned with key policy entities. The complete comparison results are available in our open-source repository.

5 Conclusion

In this study, we propose DocPolicyKG, a novel framework to construct knowledge graphs from such documents using lightweight LLMs. By integrating domain ontology, fine-tuning, and prompt engineering, DocPolicyKG significantly outperforms the base model Deepseek-R1-7B on both NER and RTE tasks, achieving performance comparable to GPT-4o. Applied to the domain of investment promotion, we build the first large-scale knowledge graph in China, covering 16,618 entities and 52,596 relations across 3,440 documents. The application of the knowledge graph is further demonstrated through Graph RAG, enhancing both the accuracy and interpretability of policy question answering by graph retrieval.

[2]https://github.com/gusye1234/nano-graphrag

6 GenAI Usage Disclosure

Prompt Design As described in the section of knowledge enhancement (Section 2.2) and prompt engineering (Section 2.4), we use Deepseek-R1 (API) to explore different prompt templates. All final prompts are designed by the authors, and any AI-generated outputs are carefully reviewed and revised to ensure correctness.

Generation of Fine-Tuning Samples We use GPT-4o (API) to generate 1,000 samples for fine-tuning on the NER tasks (Section 2.3). These samples are based on real policy document patterns but are manually reviewed, filtered, and supplemented to ensure domain fidelity and data quality.

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